

INTRODUCTION

With the rapid growth of urbanization and increasing number of vehicles, traffic congestion and rule violations have become major challenges in modern cities. Traditional traffic monitoring systems rely heavily on manual supervision, which is inefficient and prone to errors. To address these issues, intelligent traffic management systems using Artificial Intelligence (AI) and Computer Vision have emerged as effective solutions. This project proposes an AI-based traffic violation detection system that utilizes real-time video input from cameras to monitor traffic and detect violations such as red light jumping, helmet violations, and lane indiscipline. By automating the detection process and generating evidence, the system aims to enhance road safety, reduce human effort, and contribute towards smarter and more efficient traffic management.

LITERATURE SURVEY

Title	Author	Publication	Year	Key features
AI Based Traffic Violation Detection	Rakesh Bhakre, Namrata Madane, Madhur Narkhede	SMT Indira Gandhi College of Engineering	2025	• Real-time detection of traffic violations using live video feeds • AI-based system using DL and YOLO • Detects multiple violations • Automatic evidence capture • High accuracy with scalable and efficient performance
Revolutionizing Urban Mobility: A Systematic Review of AI, IoT, and Predictive Analytics in Adaptive Traffic Control System for Road Networks	Carmen Gheorghe Adrian Soica	MDPI	2025	• AI-based methods reduce traffic delays using ML techniques • Integration of IoT for real-time traffic monitoring and communication • Uses predictive analytics for congestion forecasting and traffic management • Supports intelligent features like emergency vehicle prioritization • Highlights challenges such as high cost and infrastructure requirements

Title	Author	Publication	Year	Key features
AIoT-based Smart Traffic Management System	Ahmed Mahmoud Elbasha Mohammad M.Abdellatif	British University in Cairo, Egypt	2025	•Uses AI and computer vision (YOLO) for real-time detection from CCTV footage •Calculates traffic density by counting vehicles to analyze congestion •Implements adaptive traffic control by dynamically adjusting signal timings •Cost-effective solution existing CCTV infrastructure •Improves traffic flow efficiently by ~34% compared to static systems
AI-Based Integrated Traffic Violation Detection and Smart Traffic Management System: A Comprehensive Review	Avadhut Dilip Sutar	International Journal on Science and Technology (IJSAT) – CSE	2025	•AI-based Intelligent Traffic Management Systems use computer vision, IoT, and real-time video analytics for traffic monitoring and violation detection •Deep learning models like CNN, YOLO, and Faster R-CNN enable accurate vehicle detection and classification •Detects multiple violations such as red-light jumping, helmet absence, over-speeding, and illegal parking •Integrates License Plate Recognition (LPR) and OCR for automated vehicle identification and enforcement •Improves traffic safety and efficiency but faces challenges like lighting conditions, occlusion, and scalability

EXISTING SYSTEM AND THEIR LIMITATIONS

Existing System:

Traditional traffic management systems mainly rely on fixed signal timings and manual monitoring by traffic personnel. These systems operate on predefined time intervals without considering real-time traffic conditions. In some cases, basic sensor-based systems are used to estimate traffic density, but they often lack accuracy and adaptability. Additionally, monitoring traffic violations such as signal jumping or helmet absence is largely dependent on human observation or limited camera surveillance without intelligent analysis.

Limitations:

These existing systems have several limitations. Fixed signal timings lead to inefficient traffic flow, especially during peak hours, causing congestion and increased waiting times. Manual monitoring is time-consuming and prone to human error, making it difficult to ensure consistent enforcement of traffic rules. Sensor-based systems require additional hardware and maintenance, increasing costs. Moreover, traditional systems lack real-time decision-making capabilities and cannot automatically detect and record violations, resulting in reduced efficiency in traffic management and law enforcement.

PROBLEM STATEMENT OF PROPOSED PROJECT WITH BLOCK DIAGRAM

Problem Statement:

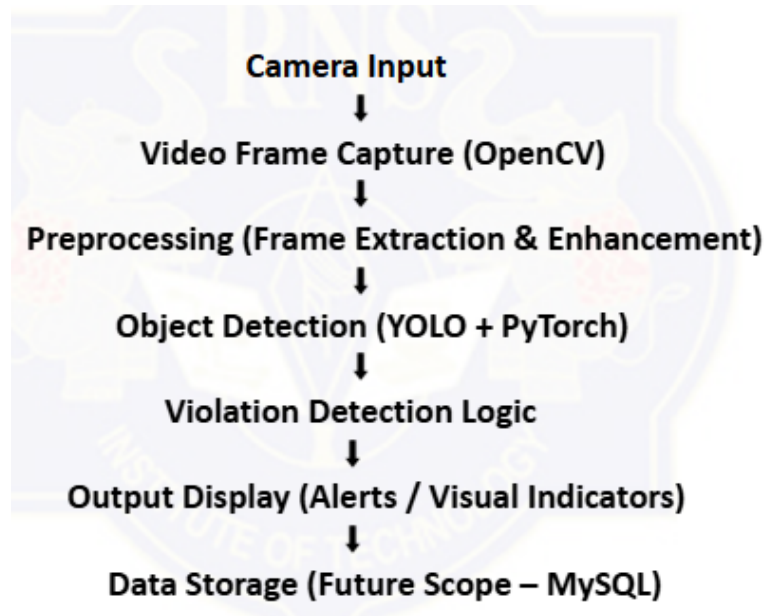
With the increasing number of vehicles in urban areas, traffic congestion and violations such as signal jumping, helmet absence, and lane indiscipline have become major concerns. Existing traffic systems lack real-time monitoring and rely heavily on manual supervision, which is inefficient and prone to errors. There is a need for an intelligent system that can automatically monitor traffic, detect violations in real time, and provide accurate evidence for enforcement. Therefore, this project proposes an AI-based traffic violation detection system that uses computer vision and real-time video processing to improve traffic management and road safety.

Block diagram:

The proposed system consists of a camera module that captures real-time traffic video. The captured video is processed using computer vision techniques where frames are extracted and analyzed. A trained AI model

(YOLO) detects vehicles and identifies traffic violations. Once a violation is detected, the system captures the evidence image along with timestamp details. The data is then stored in a database and can be displayed through a monitoring dashboard for further analysis and enforcement.

System flow-



Thus, the system provides an intelligent and automated approach to traffic monitoring by integrating real-time processing with AI-based analysis. It enhances efficiency, reduces manual effort, and contributes to the development of smart and safe transportation systems.

OBJECTIVES OF THE PROJECT

- To develop an AI-based system for real-time traffic monitoring
- To detect traffic rule violations using computer vision techniques
- To integrate hardware and software components for automated analysis
- To capture and store violation evidence such as images and timestamps
- To reduce manual monitoring and improve traffic management efficiency
- To enhance road safety through intelligent and automated systems

METHODOLOGY

Flow Diagram (Text Representation):

Camera → Video Capture → Frame Extraction → Pre-processing → Vehicle Detection (YOLO) → Violation Detection → Evidence Capture → Database Storage → Monitoring Dashboard

Detailed Methodology:

1. Video Capture

A camera module (CCTV/USB camera) is used to capture real-time traffic video from the monitoring location. The video stream serves as the primary input to the system.

2. Frame Extraction

The continuous video stream is divided into individual frames using OpenCV. These frames are processed sequentially for analysis.

3. Pre-processing

Each frame is pre-processed to improve detection accuracy. This includes resizing, noise reduction, and adjusting brightness or contrast if required.

4. Vehicle Detection using AI Model

A trained deep learning model such as YOLO is used to detect and classify vehicles (cars, bikes, buses, etc.) in each frame. The model draws bounding boxes around detected objects and labels them accordingly.

5. Traffic Analysis

The system analyzes detected vehicles to understand their movement, count, and positioning. This helps in identifying abnormal behavior or potential violations.

6. Violation Detection Algorithm

Based on predefined rules, the system detects violations such as:

- Crossing stop line during red signal
- Riding without helmet
- Lane discipline violations

The algorithm checks object position, motion, and region of interest to determine violations.

7. Evidence Capture

When a violation is detected, the system captures the corresponding frame as evidence. Additional details such as timestamp (and location, if available) are recorded.

8. Data Storage

The captured violation data (image + details) is stored in a database for future reference and analysis.

9. Monitoring Dashboard

A simple dashboard (using Flask or similar) is used to display detected violations. This allows easy monitoring and review by authorities.

10. System Optimization

The system can be further enhanced using traffic density analysis for adaptive decision making. Additionally, detection accuracy can be improved using better datasets and advanced models.

Algorithm Steps:

1. Start system
2. Capture live video stream
3. Extract frames from video
4. Pre-process each frame
5. Apply YOLO model for object detection
6. Identify vehicles and analyze movement
7. Check for violation conditions
8. If violation detected → capture image
9. Store data in database
10. Display results on dashboard
11. Repeat process continuously

HARDWARE & SOFTWARE REQUIREMENTS

SOFTWARE REQUIREMENTS:

- **Programming Language:** Python
- **Computer Vision Library:** OpenCV
- **Object Detection Model:** YOLO
- **AI/ML Framework:** PyTorch
- **Development Environment:** Visual Studio Code
- **Database (Planned):** MySQL
- **OS:** Windows

HARDWARE REQUIREMENTS:

- 1080p USB Webcam
- Tripod Stand (for stable camera placement)
- Laptop (Minimum 8GB RAM)
- Raspberry Pi 4
- Traffic Light Mini Model (for demonstration)
- Connectivity Wires

INNOVATIVENESS AND RELEVANCE OF THE PROJECT

The proposed project introduces an intelligent and automated approach to traffic monitoring by integrating Artificial Intelligence and computer vision techniques. Unlike traditional traffic systems that rely on fixed signal timings and manual supervision, this system performs real-time analysis of traffic conditions using live video feeds. The use of deep learning models such as YOLO enables accurate and efficient detection of vehicles and traffic violations.

The innovativeness of the project lies in its ability to combine multiple functionalities such as real-time monitoring, violation detection, and automated evidence capture within a single system. Additionally, the system can analyze traffic density and support intelligent decision-making, making it adaptable to dynamic traffic conditions. The integration of software and hardware components further enhances its practical applicability.

The project is highly relevant in the context of increasing urban traffic and the need for smart city solutions. It reduces human effort, improves traffic law enforcement, and enhances road safety. Moreover, The system is scalable and

can be extended for real-time deployment in smart city environments. The system can be further extended with license plate recognition and adaptive traffic signal control for enhanced automation.

ALIGNMENT WITH SDG GOALS

The proposed AI-based traffic monitoring and violation detection system aligns with the United Nations Sustainable Development Goals (SDGs), which aim to promote sustainable, safe, and efficient urban development. The project contributes to improving road safety, reducing traffic congestion, and supporting smart city infrastructure through the use of advanced technologies such as Artificial Intelligence and real-time data analysis

Aligned SDGs:

- **Good Health and Well-being:**
Reduces road accidents and improves public safety through effective traffic rule enforcement
- **Industry, Innovation, and Infrastructure:**
Promotes the use of AI and smart technologies in modern traffic management systems
- **Sustainable Cities and Communities:**
Supports development of smart cities by improving urban traffic efficiency and safety
- **Climate Action:**
Helps reduce fuel consumption and emissions by minimizing traffic congestion
- **Peace, Justice, and Strong Institutions:**
Strengthens law enforcement through automated and transparent traffic monitoring systems
- **Partnerships for the Goals:**
The system can be integrated with government systems, smart city projects, and IoT networks

EXPECTED OUTCOMES & APPLICATIONS

Expected Outcomes:

- A working prototype of an AI-based traffic violation detection system
- Accurate real-time detection of vehicles and traffic rule violations
- Automated evidence generation including images and timestamps
- Improved efficiency in traffic monitoring with reduced human intervention
- Scalable system that can be extended for real-world implementation

Applications:

- Smart traffic management systems in urban cities
- Automated monitoring at traffic signals and intersections
- Support for traffic police in law enforcement
- Accident prevention through strict rule enforcement
- Integration with smart city and intelligent transportation systems
- Traffic data analysis for planning and optimization